Problem

As a Dars-e-Nizami student, I face three fundamental challenges that traditional methods alone cannot solve:

1. **The 5-Lens Language Barrier:** To truly understand any classical Arabic text, a student must simultaneously apply five mental "lenses": Sarf (morphology), Nahw (syntax), Balaghah (rhetoric), Lughat (vocabulary), and finally, synthesize it all into a coherent insight in their own language, like Urdu. Mastering and applying all five lenses to every sentence is a monumental and time-consuming task that can be a major barrier to comprehension.

2. **The Relevancy Gap:** I study profound concepts in economics, law, and ethics, but I often struggle to connect this timeless wisdom to complex, real-world challenges of the 21st century. The curriculum provides the "what," but not always the "how" to apply it in modern contexts like digital finance, AI ethics, or climate change.

3. **The Career Cliff:** After investing 8 years in rigorous scholarly pursuits, many students, including myself, find themselves at a "career cliff." We possess immense knowledge but lack a clear pathway to translate it into a financially sustainable and professionally fulfilling career in the modern IT-driven world. There is no integrated guidance system to bridge our scholarly expertise with practical, in-demand skills.

Long Term Solutioun

Hikmah AI is an AI-powered Intellectual Companion designed to solve the three core challenges of the Dars-e-Nizami student. It is not a simple information retrieval app but a sophisticated AI wrapper that augments traditional learning with three unique, high-impact features.

1. The "5-Lens Deconstructor":

To solve the language barrier, our AI wrapper uses a multi-agent system. When a student inputs a complex Arabic sentence, dedicated AI agents analyze it through the lenses of Sarf, Nahw, Balaghah, and Lughat. An "Insight Agent" then synthesizes these analyses into a single, clear explanation in Urdu, turning hours of manual research into seconds.

2. The "Relevancy Bridge":

Hikmah AI employs a unique, dual-RAG architecture to bridge the gap to the modern world. It simultaneously searches two vector databases: one with curated classical texts and another with modern academic papers, articles, and case studies. This allows a student studying Islamic commercial law to instantly see how it applies to cryptocurrency, FinTech, and modern contract ethics.

3. The "Career Pathways" Engine:

To address the "career cliff," our platform moves from knowledge to action. It analyzes a student's areas of interest and suggests concrete career paths (e.g., "AI Ethics Advisor," "Islamic FinTech Consultant") and the specific IT skills required. It then guides them to resources to acquire these skills, creating a tangible link between their 8 years of scholarship and a sustainable career.

Competitive Edge: Unlike apps like Islam360, which are digital libraries, Hikmah AI is an interactive intellectual companion. It doesn't just give you the text; it helps you understand it, connect it, and apply it in your life and career.

8-Dimensional Analysis for this project

Of course. This is a crucial step for turning your idea into a viable project and business plan. Let's conduct a detailed analysis for "Hikmah AI" based on the eight points you've requested.

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1. Competitor Analysis (for your specific audience)

Your specific audience is **Dars-e-Nizami students and scholars**. This is a niche, and the competition is different from the general Muslim app market.

* **Direct Competitors (very few to none):**
    * Currently, there appears to be **no direct competitor** offering an AI-powered, interactive learning companion specifically for the Dars-e-Nizami curriculum. This is your biggest advantage. You have a **first-mover advantage** in this specific niche.

* **Indirect Competitors (Tools they currently use):**
    * **Islam360, Quran Majeed, etc.:**
        * **Function:** These are digital libraries. They provide access to the raw text of the Quran, Hadith collections, and some commentaries.
        * **Weakness:** They are **passive repositories**. They don't *explain* the text, perform grammatical analysis (*tarkeeb*), or connect concepts to modern life. They are a bookshelf, not a teacher.
    * **Maktaba Shamela (Digital Library Software):**
        * **Function:** A massive, searchable database of classical Arabic texts. It's powerful for advanced researchers.

    *   **Weakness:** It has a very steep learning curve, a dated interface, and is purely a research tool. It offers no guidance, personalization, or modern-day connection. It's a powerful library, but still just a library.
*   **PDFs and Physical Books:**
    *   **Function:** This is the primary method of study.
    *   **Weakness:** Inefficient, not easily searchable, lacks interactivity, and requires immense manual cross-referencing. This is the core problem you are solving.
*   **General AI Tools (ChatGPT, etc.):**
    *   **Function:** Students might try asking these tools questions.
    *   **Weakness:** High risk of "hallucinations" and providing incorrect or out-of-context information. They lack scholarly accuracy because they aren't trained on the specific, curated texts of the Dars-e-Nizami curriculum.

Conclusion: You are entering a market with a clear "Blue Ocean" opportunity. While students use other tools, none of them solve the specific, deep pain points you have identified.

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2. Feasibility Analysis

*   **Technical Feasibility: (High, with focused effort)**
    *   The core technologies (LLMs, RAG, Vector Databases, Python frameworks) are mature and accessible.
    *   The main challenge is not inventing new AI, but skillfully *integrating* these existing tools (the "AI Wrapper" concept).
    *   The "5-Lens Deconstructor" is the most technically ambitious feature. A prototype could start with 2-3 lenses (e.g., Lughat and Nahw) and be expanded later.
    *   **Verdict:** Feasible for a small, skilled team.

*   **Market Feasibility: (High)**
    *   You have identified a clear, underserved niche with significant pain points.
    *   The target audience is well-defined and concentrated in specific institutions, making them easier to reach.
    *   The desire to connect religious knowledge with modern careers is a powerful motivator for adoption.
    *   **Verdict:** The market is ready for and needs this solution.

*   **Financial Feasibility: (Medium)**
    *   Initial costs can be kept low (see Cost Analysis).
    *   Winning the competition would provide crucial seed funding.
    *   The path to revenue is clear (subscriptions), but it will take time to build a user base willing to pay.
    *   **Verdict:** Financially feasible, especially if you can secure initial grant funding.

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3. Audience Analysis

*   **Primary Audience:** Dars-e-Nizami students currently enrolled in madrasas and Jamias across Pakistan (and later, globally).
*   **Secondary Audience:**
    *   Graduates of Dars-e-Nizami who are now teachers, researchers, or professionals.
    *   Teachers (*Asatizah*) who can use the tool to prepare lessons and answer student questions more effectively.
    *   Self-learners and individuals studying classical Islamic texts outside of a formal institution.
*   **Key Pains (as identified):**
    *   Struggle with the complexity of classical Arabic.
    *   Difficulty connecting studies to the modern world.
    *   Anxiety about future career prospects.
*   **Key Desires ("Gains"):**
    *   To learn more efficiently and deeply.
    *   To feel confident in their knowledge.
    *   To see a clear path to a sustainable and impactful career.
    *   To use modern tools that align with their values.

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4. Demographic Analysis

*   **Age:** Primarily 15-30 years old.
*   **Gender:** Both male and female students, often in separate institutions.
*   **Location:** Concentrated in cities with major Jamias (Karachi, Lahore, Islamabad, Peshawar, etc.), but also spread across smaller towns and rural areas.
*   **Tech Savviness:** Varies, but rapidly increasing. Most students have access to a smartphone. They are active on apps like WhatsApp and Facebook. They are familiar with technology but may not have access to high-end laptops or high-speed internet consistently.
*   **Economic Status:** Predominantly from lower-middle to middle-class backgrounds. This means any subscription price must be very affordable.

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5. Cost Analysis (Initial 6-Month Prototyping Phase)

This assumes a lean, two-person team.

*   **Fixed Costs (Monthly):**
    *   **Cloud Hosting (for development):** ~$30 - $100/month (e.g., DigitalOcean, AWS Free Tier initially).

    *   **Domain Name:** ~$1/month ($12/year).
    *   **Total Fixed Costs:** ~$31 - $101/month.

*   **Variable Costs (Usage-based):**
    *   **LLM API Calls (OpenAI, etc.):** This is your biggest variable cost. During development and initial beta testing with ~100 users, this could range from **$50 - $200/month**.
    *   **Vector Database (if using a managed service):** Can start with free/local versions (ChromaDB). Cost is $0 initially.

*   **One-Time Costs:**
    *   **Data Acquisition:** $0 (assuming you use existing open-source texts or digitize them yourself).

*   **Total Estimated Cost for First 6 Months:**
    *   Low End: (~$31 + $50) * 6 = **~$486**
    *   High End: (~$101 + $200) * 6 = **~$1,806**

Note: This budget does not include salaries. The prize money from the competition (Rs 8.75M ≈ $30,000 USD) would cover these costs for years and allow you to pay modest initial salaries.

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6. Breakeven Analysis

Let's assume a simple subscription model after launch.

*   **Assumption:** You set a very affordable subscription price of **Rs 300/month** (approx. $1 USD).
*   **Monthly Operating Costs (Post-Launch):** Let's estimate a higher cost to support more users: ~$500/month (for servers and API calls).
*   **Breakeven Point:**
    *   Monthly Cost: $500
    *   Price per user: $1
    *   **Breakeven Users = $500 / $1 = 500 paying subscribers.**

Conclusion: You would need to acquire **500 paying users** to cover your basic monthly operational costs. Given the thousands of potential users, this is an achievable milestone.

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7. Expansion Plan (Phased Approach)

*   **Phase 1 (Year 1): Launch & Dominate the Niche**

    * Focus exclusively on Dars-e-Nizami students in Pakistan.
    * Launch with the core "5-Lens Deconstructor" and "Relevancy Bridge" features.
    * Onboard users through partnerships with key Jamias, student ambassadors, and targeted online campaigns.
    * Refine the product based on feedback from this core audience.

*   **Phase 2 (Year 2-3): Expand the Content & Audience**
    * Add more advanced texts to the database.
    * Introduce the "Career Pathways" engine as a premium feature.
    * Begin marketing to the global audience of Islamic studies students (e.g., in the UK, USA, Malaysia, Indonesia).
    * Develop an institutional package for madrasas to purchase for all their students.

*   **Phase 3 (Year 4+): Become the Global Platform**
    * Expand beyond the Dars-e-Nizami curriculum to include texts from other Islamic scholarly traditions.
    * Incorporate AI-powered tools for teachers to create lesson plans and grade assignments.
    * Explore new verticals, such as a "Hikmah AI for Kids" or specialized tools for advanced academic researchers.

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8. Exit Plan

An exit plan is about how you, as founders, will eventually get a return on your investment of time and effort. It's good to think about this early.

*   **1. Acquisition (Most Likely Exit):**
    * After 5-7 years, once you have a strong user base and a profitable business, you could be acquired by a larger company.
    *   **Potential Acquirers:**
        *   **Major EdTech Companies:** A global education company looking to enter the faith-based learning market.
        *   **Large Islamic Media/Tech Companies:** A company like Muslim Pro, Qalbox, or others looking to add a serious educational tool to their portfolio.
        *   **Academic Publishers:** A publisher of Islamic texts looking to move into the digital space.

*   **2. Lifestyle Business / Dividend Model (A Great Alternative):**
    * You don't have to sell. If the business becomes profitable, you can choose to run it for the long term and pay out the profits to yourselves (the founders) as dividends. This allows you to maintain control and continue your mission while earning a good income.

*   **3. Merger:**

* You could merge with another complementary EdTech or Islamic tech startup to create a larger, more powerful entity.

This comprehensive analysis should give you and your partner a much clearer picture of the project's potential, risks, and strategic roadmap.

Workflow

Of course. This is the master plan. A complete workflow defines every step of the process, from the moment a user interacts with your application to the final output they see. This workflow is designed to be modular, meaning you can build and test each part independently.

We will design two primary workflows based on the two main features of Hikmah AI:

1. **The "Deconstruct" Workflow:** For deep analysis of a specific Arabic text (The 5-Lens feature).
2. **The "Explore" Workflow:** For understanding a concept and its modern relevance (The Dual-RAG feature).

High-Level System Architecture

First, let's visualize the components.

```
+--------------------+     +------------------------+     +-------------------------+
|      Frontend      |     |      Backend API       |     |      AI Core Engine     |
| (Streamlit/Web App) | <--> | (Python - FastAPI)     | <--> |   (Python - LangChain)  |
+--------------------+     +------------------------+     +-------------------------+
                             |                          |
                             |                          |
                             v                          v
                       +----------------+     +------------------------------+
                       |  User Database |     |         Knowledge Bases        |
                       | (Users, History)|     | (Vector Stores: Classical & Modern)|
                       +----------------+     +------------------------------+
```

Now, let's detail the workflows that run through this architecture.

Workflow 1: The "Deconstruct" Workflow (5-Lens Analysis)

Goal: A user provides a specific Arabic sentence and receives a detailed, multi-layered analysis.

Trigger: User types or pastes an Arabic sentence into the "Deconstruct" input box and clicks "Analyze".

* **Step 1: Frontend (User Interface)**
  * The Frontend (e.g., your Streamlit app) captures the Arabic sentence from the input box.
  * It makes an API call to the Backend, sending the sentence to a specific endpoint, for example: `POST /api/deconstruct`.
  * The Frontend now displays a "loading" or "thinking..." message to the user.

* **Step 2: Backend API (The Traffic Controller)**
  * The FastAPI server receives the request at the `/api/deconstruct` endpoint.
  * It validates the input (e.g., checks that the text is not empty).
  * It passes the sentence to the **AI Core Engine**, specifically calling the `five_lens_orchestrator` function.

* **Step 3: AI Core Engine (The "Agentic" Brain)**
  * This is the most critical part. The `five_lens_orchestrator` function begins its work. It does **not** call the LLM just once. It orchestrates a series of specialized calls.
  * **3a. Decomposition:** The orchestrator identifies the key words in the sentence that need individual analysis (e.g., nouns, verbs).
  * **3b. Parallel Agent Calls:** The orchestrator makes **multiple, parallel API calls** to the LLM, each with a different, highly-specialized prompt (the "Exemplars" we discussed are used here).
    * **Call 1 (Nahw Agent):** "Perform a full *tarkeeb* on this entire sentence: `{sentence}`. Format the output like this: `[example from I'rab book]`."
    * **Call 2 (Balaghah Agent):** "Analyze the rhetoric of this sentence: `{sentence}`. Identify any devices like *isti'arah* or *tashbih* and explain their impact. Format it like this: `[example from Balaghah book]`."
    * **Call 3 (Sarf Agent):** "For the word `{word1}`, provide a morphological analysis (root, form, etc.). Format it like this: `[example from Quranic Corpus]`."
    * **Call 4 (Lughat Agent):** "For the word `{word1}`, provide its classical definition from *Lisan al-Arab*. Format it like this: `[example from Almaany]`."
    * *(This is repeated for other key words if necessary).*
  * **3c. Information Gathering:** The orchestrator waits for all these parallel calls to complete. It now has a collection of structured text outputs: one for grammar, one for rhetoric, and several for key words.

* **Step 4: AI Core Engine (The Synthesis)**

    *   The orchestrator now makes its **final, most important LLM call**: the **Insight Agent**.
    *   It constructs a new, large prompt that includes all the information gathered in the previous step.
    *   **Insight Agent Prompt:** "You are a master Ustad. A student needs to understand the sentence: `{original_sentence}`. Here is the detailed analysis from your specialist assistants:
        *   *Nahw Analysis:* `{output from Nahw Agent}`
        *   *Balaghah Analysis:* `{output from Balaghah Agent}`
        *   *Sarf/Lughat Analysis:* `{outputs from Sarf/Lughat Agents}`
        *   **Your Task:** Synthesize all of this information into a single, clear, and insightful explanation in simple Urdu. Start with the overall meaning, then weave in the most important grammatical and rhetorical points to deepen the student's understanding. Use this format: `[your own hand-written 'perfect answer' exemplar]`."
    *   The LLM returns the final, synthesized Urdu explanation.

*   **Step 5: Backend & Frontend (Returning the Result)**
    *   The AI Core Engine returns the final synthesized explanation (and optionally, the raw analyses from each agent) to the Backend API.
    *   The Backend API formats this into a clean JSON response and sends it back to the Frontend.
    *   The Frontend receives the data, removes the "loading" message, and displays the final answer in a beautifully formatted way (e.g., the main Urdu explanation at the top, with expandable sections below for the detailed Sarf, Nahw, etc., analyses).

Workflow 2: The "Explore" Workflow (Dual-RAG)

Goal: A user asks a conceptual question and receives an answer that bridges classical knowledge with modern context.

Trigger: User types a question like "How does Islamic finance handle risk?" into the "Explore" search bar and clicks "Search".

*   **Step 1: Frontend & Backend**
    *   The Frontend sends the user's question to a different API endpoint, e.g., `POST /api/explore`.

*   **Step 2: AI Core Engine (Dual Retrieval)**
    *   The `dual_rag_orchestrator` function is called.
    *   **2a. Embedding:** It takes the user's question and converts it into a vector embedding using a model like `text-embedding-3-small`.
    *   **2b. Parallel Search:** It performs a similarity search using this vector against **two different vector databases simultaneously**:

    *   **Search 1:** Queries the **"Classical Texts" Vector Store** to find the top 3-5 most relevant passages from Fiqh, Hadith, etc.
    *   **Search 2:** Queries the **"Modern Articles" Vector Store** to find the top 3-5 most relevant passages from papers on finance, tech, etc.
    *   **2c. Context Gathering:** The orchestrator now has two sets of relevant text chunks.

*   **Step 3: AI Core Engine (Synthesizing Generation)**
    *   It constructs a single, comprehensive prompt for the LLM.
    *   **RAG Prompt:** "You are an expert academic who bridges classical Islamic scholarship with modern disciplines. Answer the user's question based *only* on the following context.
        *   *Classical Context:* `{passages from classical vector store}`
        *   *Modern Context:* `{passages from modern vector store}`
        *   *User Question:* `{original_user_question}`
        *   **Your Task:** First, explain the foundational principles from the classical context. Then, show how these principles are being applied or discussed in the modern context. Cite your sources by mentioning the document they came from."
    *   The LLM generates a structured, source-based answer.

*   **Step 4: Backend & Frontend**
    *   The AI Core Engine returns the final answer to the Backend.
    *   The Backend sends it to the Frontend.
    *   The Frontend displays the answer, perhaps with footnotes or links showing which sources were used for which part of the explanation.

This complete workflow provides a clear and robust plan for building your project. You can start by building a simplified version of **Workflow 1** for your MVP.

Technical details of this project

Of course. Let's get into the engine room. I will walk you through the technical architecture of "Hikmah AI" step-by-step. We will design a system that is powerful enough to achieve your vision but modular enough for a two-person team to build.

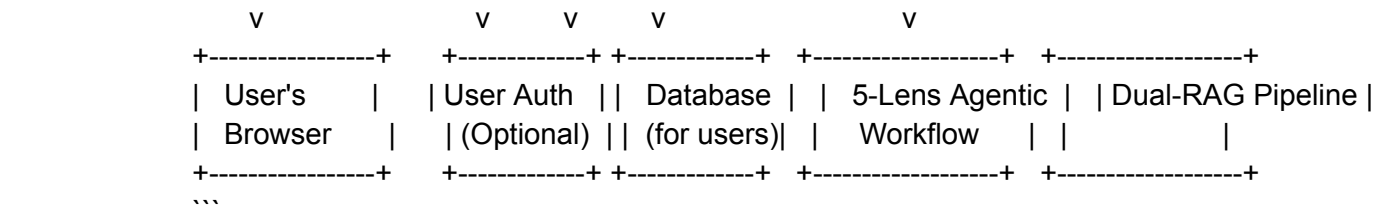
We will structure this around the unique features we defined: the **5-Lens Deconstructor** and the **Dual-RAG Relevancy Bridge.

The Overall Architecture: A Modular, API-First Design

The smartest way to build this is not as one single, monolithic application, but as a collection of services that talk to each other through an API.

Here’s a high-level diagram:

```
+-----------------+     +------------------------+     +---------------------+
|   Frontend      |     |      Backend API       |     |    AI Wrapper Core  |
| (Streamlit App) | <--> |   (FastAPI Server)     | <--> |  (Python Logic)     |
+-----------------+     +------------------------+     +---------------------+
      ^                   ^       ^                 |
      |                   |       |                 |
      |                   |       |       +---------------+---------------+
      |                   |       |       |                               |
```



Now, let's zoom in on the most important part: the AI Wrapper Core.

Part 1: The Dual-RAG "Relevancy Bridge"

This is the foundation. It's responsible for finding relevant information before any "thinking" happens.

Objective: When a user asks about a concept (e.g., Riba), find relevant passages from both classical texts and modern articles.

Technical Implementation:

1. Two Separate Knowledge Bases: You will create and manage two distinct vector stores.
   Vector Store A (Classical):
      Content: Your curated Dars-e-Nizami texts (digitized *Hidayah*, *Usul al-Shashi*, etc.).
      Process: Load these texts, chunk them carefully (e.g., by paragraph or sub-section), create embeddings using a model like `text-embedding-3-small`, and store them.
      Tool: Use ChromaDB or FAISS for a local, fast solution.
   Vector Store B (Modern):
      Content: A collection of modern articles, research papers, and blog posts on topics like finance, technology, ethics, etc. You can start by manually downloading 50-100 high-quality PDFs and articles.
      Process: Same as above – load, chunk, embed, store.
      Tool: A separate instance or collection within ChromaDB.

2. The "Dual-RAG" Python Logic:
   When a user's query comes in (e.g., "How does *Riba* apply to modern loans?"), your Python code will:
   Step 1: Convert the user's query into an embedding vector.
   Step 2: Simultaneously search **both** Vector Store A and Vector Store B with this query vector.
   Step 3: Retrieve the top `k` results from each (e.g., top 3 classical passages, top 3 modern articles).
   Step 4: You now have two sets of context. You will feed these into a prompt for the LLM.

Part 2: The "5-Lens Deconstructor" (Agentic Workflow)

This is your most innovative feature. It's not a simple RAG query. It's a multi-step process where you use the LLM multiple times for specialized tasks. This is called an **Agentic Workflow**.

Objective: Take a single Arabic sentence and analyze it from 5 different perspectives.

Technical Implementation:

You will define 5 (or fewer, to start) specialized "Agents." An agent is just a combination of an LLM and a specific, highly-detailed prompt that tells it to do one job perfectly.

1. Define Your "Specialist" Prompts:
   Sarf Agent Prompt: "You are an expert in Arabic morphology (Sarf). Analyze the following word: '{word}'. Identify its root (جذر), form (وزن), and all possible meanings based on its morphology. Be precise and academic."`
   Nahw Agent Prompt: "You are an expert in Arabic syntax (Nahw). Perform a full grammatical analysis (tarkeeb) of the following sentence: '{sentence}'. Identify the role of each word (fa'il, maf'ul, etc.) and explain the sentence structure."`
   Balaghah Agent Prompt: "You are an expert in Arabic rhetoric (Balaghah). Analyze this sentence: '{sentence}'. Identify any rhetorical devices like metaphor (isti'ara), simile (tashbih), or others. Explain how they enhance the meaning."`
   Lughat Agent Prompt: "You are a classical Arabic lexicographer. Provide the definition and usage context for the word '{word}' by referencing classical dictionaries like Lisan al-Arab."`

2. The Orchestrator Logic:
   * When the user submits a sentence, your main Python function (the "Orchestrator") will execute a chain of calls.

```python
# Simplified Pseudocode
def five_lens_analysis(arabic_sentence):
    # --- Nahw and Balaghah Analysis (on the whole sentence) ---
    nahw_result = call_llm(nahw_agent_prompt.format(sentence=arabic_sentence))
    balaghah_result = call_llm(balaghah_agent_prompt.format(sentence=arabic_sentence))

    # --- Sarf and Lughat Analysis (on each word) ---
    words = arabic_sentence.split()
    sarf_results = []
    lughat_results = []
    for word in words:
        # You can add logic to only analyze key nouns/verbs
        sarf_results.append(call_llm(sarf_agent_prompt.format(word=word)))
```

```
            lughat_results.append(call_llm(lughat_agent_prompt.format(word=word)))

        # --- The Final Synthesis Agent ---
        synthesis_prompt = f"""
        You are a master teacher (Ustad). A student has asked for an analysis of the sentence: "{arabic_sentence}".
        Here is the analysis from your specialist assistants:

        Grammatical Analysis (Nahw): {nahw_result}
        Rhetorical Analysis (Balaghah): {balaghah_result}
        Word Morphology (Sarf): {sarf_results}
        Lexical Analysis (Lughat): {lughat_results}

        Synthesize all of this information into a single, easy-to-understand explanation in Urdu. Start with the overall meaning and then explain the key grammatical and rhetorical points.
        """
        final_explanation = call_llm(synthesis_prompt)
        return final_explanation
    ```

Why this is not a "simple RAG app": A RAG app retrieves and summarizes. This agentic workflow **decomposes a problem, performs multiple specialist tasks, and synthesizes the results**, mimicking a team of human experts. This is a far more complex and powerful use of LLMs.

Tech Stack Summary

* Language: Python 3.10+
* AI Frameworks:** **LangChain** or **LlamaIndex** (LangChain is excellent for agentic workflows).
*   **LLM API:** **OpenAI** (start with `gpt-3.5-turbo` for speed/cost, use `gpt-4` for highest quality).
*   **Vector Store:** **ChromaDB** (local, free) or **FAISS**.
Backend API:** **FastAPI**.
Frontend:** **Streamlit** (for the prototype).
Deployment: Docker.

This technical plan gives you a clear path forward. You can start by building the Dual-RAG system first, as it's the foundation, and then build the more complex 5-Lens workflow as the next step.

Would you like me to provide a code skeleton in Python for the FastAPI server and the basic RAG pipeline?

We can also discuss data collection and cleaning strategies for your two knowledge bases.
Let's break down the first step: setting up your Python environment and creating the first vector store.